Tommaso Ballarin

ballarintom@gmail.com | LinkedIn | Portfolio

Work Experience

Artificial Intelligence Engineer, CogniteX — September 2026 - Present

- Computer Vision, Machine Learning, Deep Learning

Intern, Euclid Labs — January 2026 - April 2026

- Programming in C# and KRL (Kuka Robot Language)
- Hands-on experience with KUKA and SCARA robots: path planning, bin picking, singularity analysis

Education

Computer Engineering – Università degli Studi di Padova — Expected December 2026
Master's Degree in Computer Engineering (AI and Robotics), taught in English

Biomedical Engineering - Università degli Studi di Padova — September 2024
Bachelor's Degree in Biomedical Engineering

Projects

Visual Anomaly Detection of Varroa Destructor in Beehives

- Design and implementation of an advanced computer vision pipeline for beehive health monitoring. The system combines detection and instance segmentation models (YOLO and SAM) to isolate individual specimens within complex frames, followed by unsupervised Visual Anomaly Detection enabling the detection of anomalies and parasites at both local and global level.
- PyTorch, YOLO, Segment Anything (SAM), OpenCV, Visual Anomaly Detection, Instance Segmentation

Deep Visual Attention Prediction via Swin-Transformers — tommyballa/Visual-Saliency-Prediction.git

- Development of a Deep Learning architecture for Saliency Prediction based on Swin-Transformer, optimized for extracting complex spatial features through Computer Vision techniques.
- Skills: PyTorch, Google Colab, Python

Skills

Programming languages: Python (PyTorch, numpy), C++ (OpenCV, ROS2), SQL, Matlab
Languages: Italian, English C1
Soft skills: problem solving, teamwork, excellent time management

Personal Information

Address: via Londra 11, Vigonza (PD), Italy
Phone: +39 3451299821
Mail: ballarintom@gmail.com